共射

(1) 电路及晶体管 2N222 $\beta=120$　　已知: $V_{CC}=+12V$ $R_L=3k\Omega$ $V_i=10mV$ $R_S=600\Omega$

(2) 静态工作点: R_E, R_{B1}, R_{B2}　　要求: $|A_V|>15V/V$ $R_i>3k\Omega$ $f_L<50Hz$

取 $I_C=1mA$

取 $V_B=\frac{1}{4}V_{CC}=3V \Rightarrow R_E=\frac{V_B-V_{BE}}{I_C}=\frac{3-0.7}{1}=2.3k\Omega$

当 $I_{R_1} \gg I_B$ 时, $V_{BB}\approx V_B=3V$

$\Rightarrow 12\times\frac{R_{B2}}{R_{B1}+R_{B2}}=3 \Rightarrow R_{B1}:R_{B2}=3:1$

① $R_i>3k\Omega$: 取 $R_{B2}>3R_i$下限 $\Rightarrow$ 取 $R_{B2}=15k\Omega$ $R_{B1}=45k\Omega$

② $I_{RB_1}\gg I_B$: 取 $I_{RB_1}=10I_B=0.083mA$ $\Rightarrow$ 取 $R_{B1}=\frac{V_{CC}-V_B}{I_{R_1}}=108k\Omega$ $R_{B2}=36k\Omega$

综上, 取标称值 $R_{B2}=20k\Omega$ $R_{B1}=60k\Omega$

↳ $36k\Omega$ 固定 + $50k\Omega$ 可调

(3) 动态参数: R_C, R_{E1}

取 $I_C=1mA \Rightarrow r_e=\frac{V_T}{I_C}=26\Omega$

取 $|A_V|=20V/V \Rightarrow \frac{R_C\|3000}{26+R_{E1}}=20$

① $R_i>3k\Omega$: 取 $R_i=6k\Omega$ $\Rightarrow R_{E1}=56.6\Omega \Rightarrow R_C=3.68k\Omega$

$\Rightarrow R_{E2}=R_E-R_{E1}=2.243k\Omega$

② 输出摆幅: 取 $V_C=\frac{2}{3}V_{CC}=8V \Rightarrow R_C=\frac{V_{CC}-V_C}{I_C}=4k\Omega \Rightarrow R_{E1}=60\Omega \Rightarrow R_{E2}=2.24k\Omega$

综上, 取标称值 $R_C=3.9k\Omega$ $R_{E1}=56\Omega$ $R_{E2}=2.2k\Omega$

(4) 电容　取 $f_L=40Hz$

$$\Rightarrow C_E=\frac{1}{2\pi f_L\left[R_{E2}\|\left(R_{E1}+r_e+\frac{R_S\|R_1\|R_2}{1+\beta}\right)\right]}\approx 45\mu F$$

取标称值 $C_E=47\mu F$ $C_1=C_2=22\mu F$

验算

$V_{BB}=V_{CC}\frac{R_{B2}}{R_{B1}+R_{B2}}$ $R_{BB}=R_{B1}\|R_{B2}$

$I_E=\frac{V_B-V_{BE}}{R_E+\frac{R_{BB}}{1+\beta}}$

$V_{CE}=V_{CC}-I_CR_C-I_E(R_{E1}+R_{E2})$

$V_B=\frac{(1+\beta)(r_e+R_E)}{(1+\beta)(r_e+R_E)+R_{BB}}$

$I_C=\alpha I_E\approx 1mA$ $V_C=V_{CC}-I_CR_C>V_B-0.4$

$r_e=\frac{V_T}{I_C}$

$V_{BE}=V_B-I_E(R_{E1}+R_{E2})$

$A_v=-\frac{R_C\|R_L}{r_e+R_{E1}}>15V/V$

当$R_L=\infty$时

$A_v=-\frac{R_C}{r_e+R_{E1}}$

$R_{in}=R_{B1}\|R_{B2}\|[(1+\beta)(r_e+R_{E1})]>3k\Omega$

$R_{out}\approx R_C=4.3k\Omega$

$f_L<50Hz$

实验

$R_i=\frac{V_i}{V_s-V_i}\times R$

$R_o=\frac{V_o-V_o'}{V_o'}\times R_L$

反相放大

要求：$R_i\geq 15k\Omega$ $|A_v|=15V/V$

$A_v=-\frac{R_F}{R_1}$

$R_i=R_1$

$R_o=0$

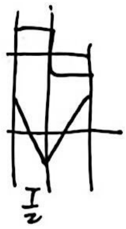
取$R_p=R_1\|R_F$

反相积分器

$V_o(t)=-\frac{1}{R_1C}\int_0^t V_i(t)dt$

$V_i(t)=\begin{cases}0 & t<0\\ E & t>0\end{cases}\rightarrow A$

$V_o(t)=-\frac{E}{R_1C}t\rightarrow B$

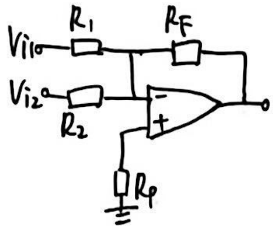


$\Rightarrow 2B=\frac{A}{R_1C}\frac{T}{2}$

C=0.022uF 取$R_f\gg\frac{1}{j\omega C}=\frac{1}{2\pi fC}$ $\omega=2\pi f$

$\Rightarrow R_f\gg\frac{2B}{\pi A}R_1$ 取$R_f>10R_1$

算术运放



取$R_p=R_1\|R_2\|R_F$

当取$R_1=R_2$ $R_F=R_p$时 $\begin{cases}V_{i2}:+\\ V_{i1}:-\end{cases}$

$V_o=\frac{R_F}{R_1}(V_{i2}-V_{i1})$

否则

$V_o=\left(\frac{R_p}{R_2+R_p}\frac{R_1+R_F}{R_1}V_{i2}-\frac{R_F}{R_1}V_{i1}\right)$

迟滞比较器

$V_6=V_{OH}$时 $V_+'=\frac{R_{43}}{R_{43}+R_{44}}(V_{Z2}+V_{D1})+\frac{R_{44}}{R_{43}+R_{44}}V_{42i}$

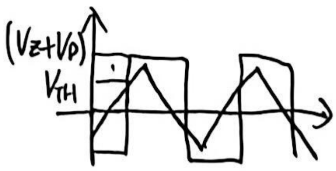
$V_6=V_{OL}$时 $V_+''=-\frac{R_{43}}{R_{43}+R_{44}}(V_{Z1}+V_{D2})+\frac{R_{44}}{R_{43}+R_{44}}V_{42i}$

当$V_6=V_{OH}$时

要使 $\begin{cases}V_{OH}时, V_+'>0 \Rightarrow V_{42i}>-\frac{R_{43}}{R_{44}}(V_Z+V_D)=-V_{TH}\\ V_{OL}时, V_+'<0 \Rightarrow V_{42i}<-V_{TH}\end{cases}$

当$V_6=V_{OL}$时

要使 $\begin{cases}V_{OL}, V_+''<0 \Rightarrow V_{42i}<V_{TH}\\ V_{OH}, V_+''>0 \Rightarrow V_{42i}>V_{TH}\end{cases}$



正相

$V_{TH}=\frac{R_{43}}{R_{44}}(V_Z+V_D)$